Network Monitoring With AMP

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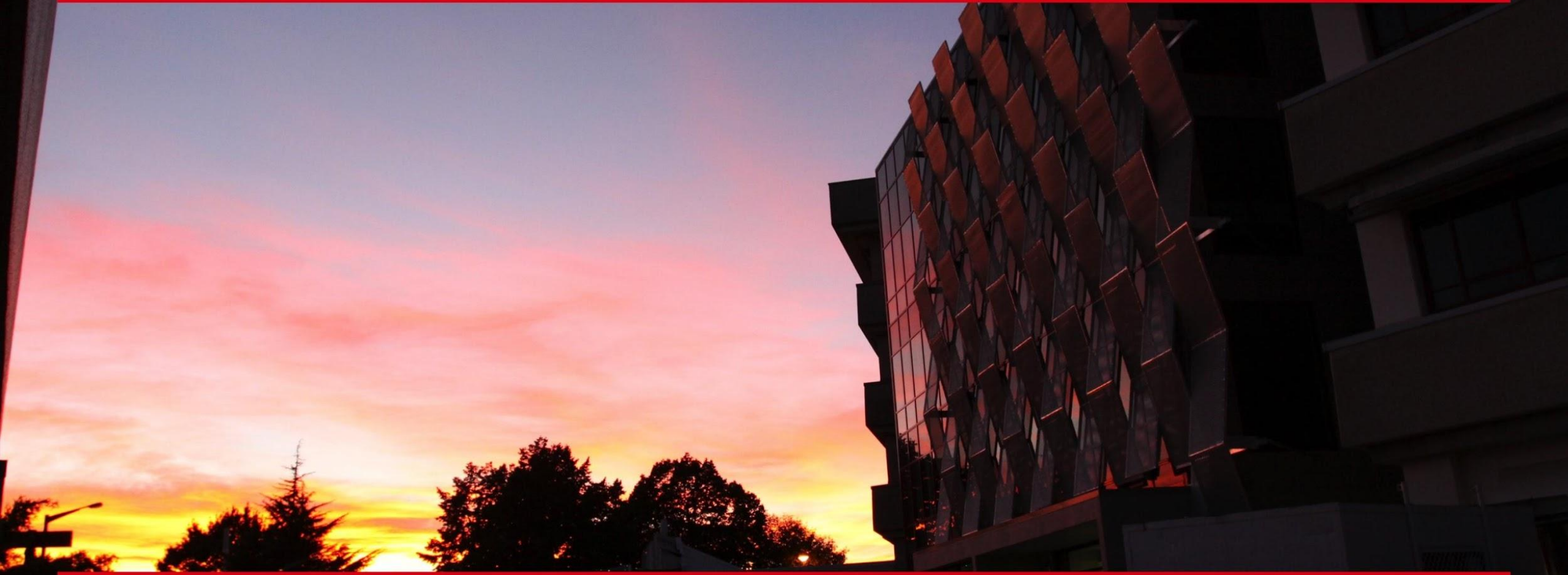


Active Measurement Project

What Is Active Network Measurement?

- Putting packets onto the network, measuring how they behave
  - e.g. `ping`
- Usually measuring end-to-end and/or endpoint characteristics
  - How quickly?
  - How much?
  - Who answers?
  - What did they say?
- Can interfere with “real” traffic on the network
  - And our own measurements too if we aren’t careful

What Is AMP?

- Continuous, distributed active network measurement
- Data exploration
  - All the data is available on the website to browse and compare
- Event detection
  - To save actually having to look at ALL the data

Modern Day AMP

- Originally measured NSF HPC networks in the USA
- New version written over the last few years using lessons learnt
  - Dependencies aren't always evil
  - Processes vs threads
- Added tests to infrastructure targets outside our control
  - DNS, HTTP, etc
- Public deployment across NZ ISPs and universities
- Private internal deployments inside other organisations

Current AMP Tests

- Latency measurements
  - ICMP, TCP, UDP, DNS
- UDP traceroute
- TCP upload and download throughput
- HTTP page download
- YouTube video streaming

The User Experience

- Software and the network can interact in unpredictable ways
  - Measuring behaviour in isolation might not tell the full story
- Use the same software as the user, or the same techniques
  - UDP latency test can send regularly spaced packets (like VOIP)
  - HTTP test uses multiple parallel connections, pipelining
  - YouTube test uses Chromium
- Bits and milliseconds aren't always relatable
  - What do the numbers mean to the average user?

YouTube Test

- Use Chromium to download and play a YouTube video
- Behaves just like it would for a normal user
  - Downloads from whatever server Google tells us to
  - Only downloads enough to play smoothly
- Report interesting statistics
  - How long the initial buffering takes
  - How often buffering occurs
  - What resolution changes occur

YouTube Test

- Simple webpage that loads a video using the IFrame API
  - https://developers.google.com/youtube/iframe_api_reference
- Measure the time between different events
  - State change (buffering, playing, finished, etc)
  - Video quality change
- YouTube test loads page in a headless Chromium application
- Fetch measurements using Chrome DevTools protocol
  - Needed to get the data from webpage JavaScript into C++ application

YouTube Test

- Why not just run headless Chromium as-is and use DevTools?
  - I want to use the same approach to make a new HTTP test
- Performance Navigation Timing has lots of timing information
  - Same information used to make “waterfall” diagrams
- Cross-origin resources need `Timing-Allow-Origin` header
  - Possible privacy issue if websites can check timing of all resources
  - Instantiating the browser in my own code lets me modify networking without having to recompile Chromium
  - I think I can force the header on all resources I download (but haven’t got it working yet)

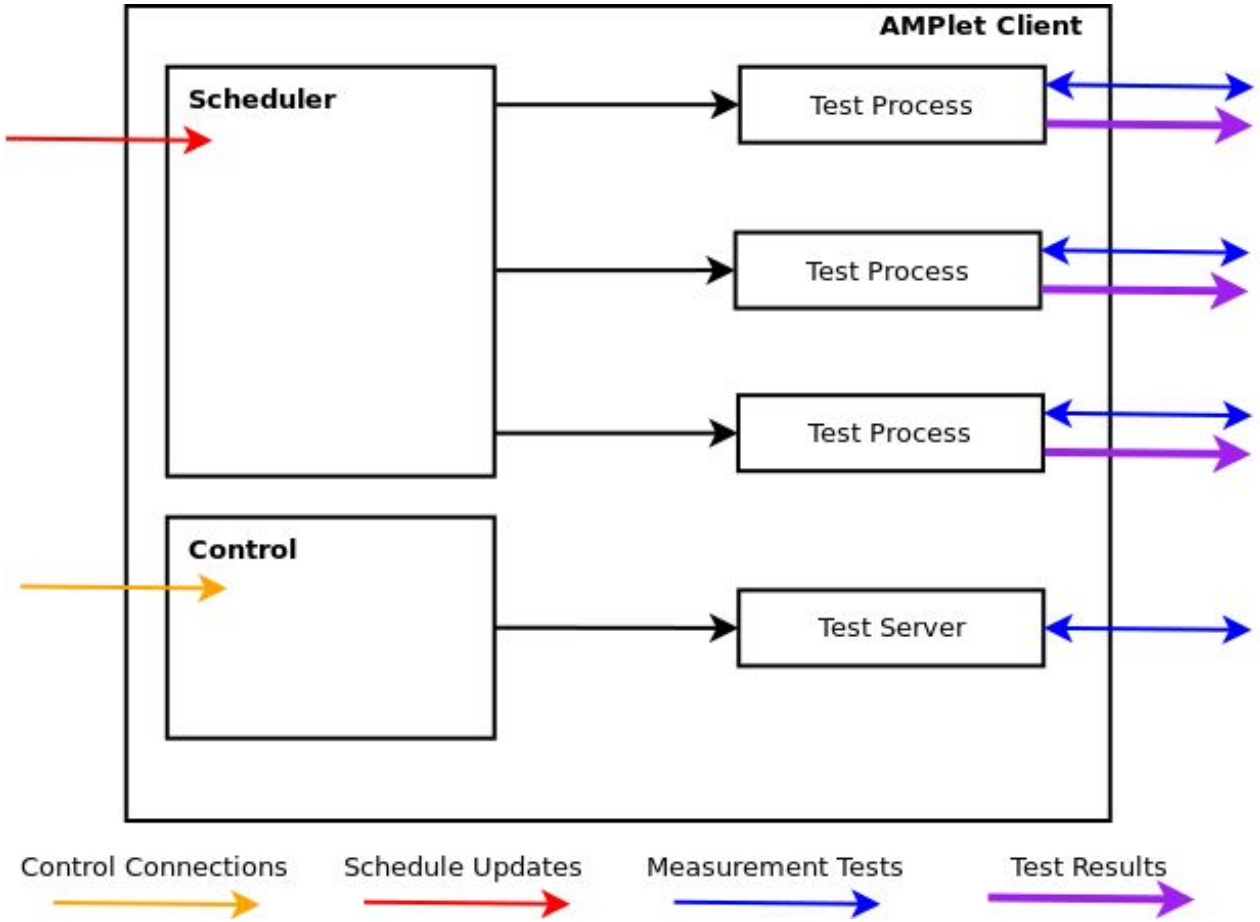
YouTube Test Problems

- Chromium is big, and changes quickly
  - Bug fixes appear quickly, hard to apply as everything else changes too
  - Shared libraries come and go (or change name) all the time
- Headless is new too and still being actively developed
  - Some symbols I needed were in undocumented object files
  - Later they became real shared libraries
- Anchor myself to the versions packaged by Debian/Ubuntu
  - Take advantage of their patches and testing

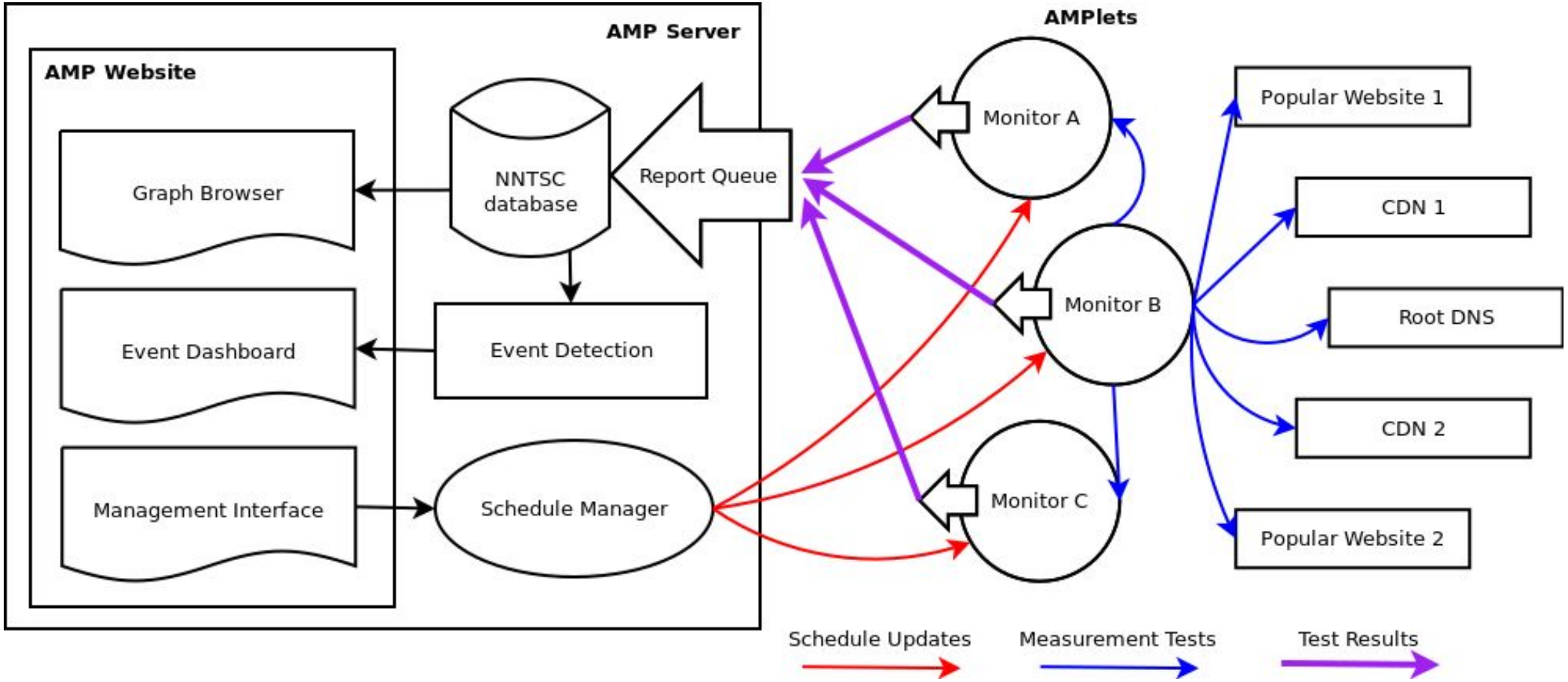
YouTube Test Problems

- Using both libssl and libboringssl together breaks things
  - Not initially, but at some point later during execution
- Chromium forks itself after performing startup duties
  - Quicker startup for future processes
  - Keeps reference to correct version of the binary
  - Fine if chromium is the original binary, not if it's amplet2
- Split behaviour across multiple programs
  - One part understands reporting AMP test data
  - Another runs Chromium and collects the data

AMP Client Architecture



AMP Architecture



Event Detection

- Network measurements are normally time series data
  - Plenty of techniques for finding anomalies in time series data
- Not many of these techniques work well in real time
  - Trade off between accuracy and timeliness
- No one perfect technique
  - Detecting increases vs decreases
  - Noisy data vs consistent data
- Data fusion
  - If many detectors fire, then it is probably an event

AMP Web Interface

EVENT DASHBOARD | GRAPH BROWSER | MATRIX | LOGIN

WAND Active Measurement Project

Show Absolute Latency from New Zealand Amplets to CDNs

a248.e.akamai.net
appldnld.apple.com
cachefly.cachefly.net
cdn-fastly.deb.debian.org
client.dropbox.com
cloud.google.com
download.microsoft.com
gcm.googleapis.com
itunes.apple.com
pbs.twimg.com
secure-appldnld.apple.com
store.steampowered.com
v1.lscache1.c.youtube.com
cloudflare.com
google-analytics.com
googleapis.com

callplus
canterbury
catalyst
csotago
inspire
lightwire
massey-pn
ns2b
ns3a
ns3b
ns4b
snap-akl
vocus
waikato

ICMP Latency

Show IPv4 only

IPv4
IPv6

Recent latency

Not tested
Missing data
Latency < 5ms
Latency 5 - 25ms
Latency 25 - 50ms
Latency 50 - 100ms
Latency 100 - 200ms
Latency 200 - 300ms
Latency >= 300ms

AMP Web Interface

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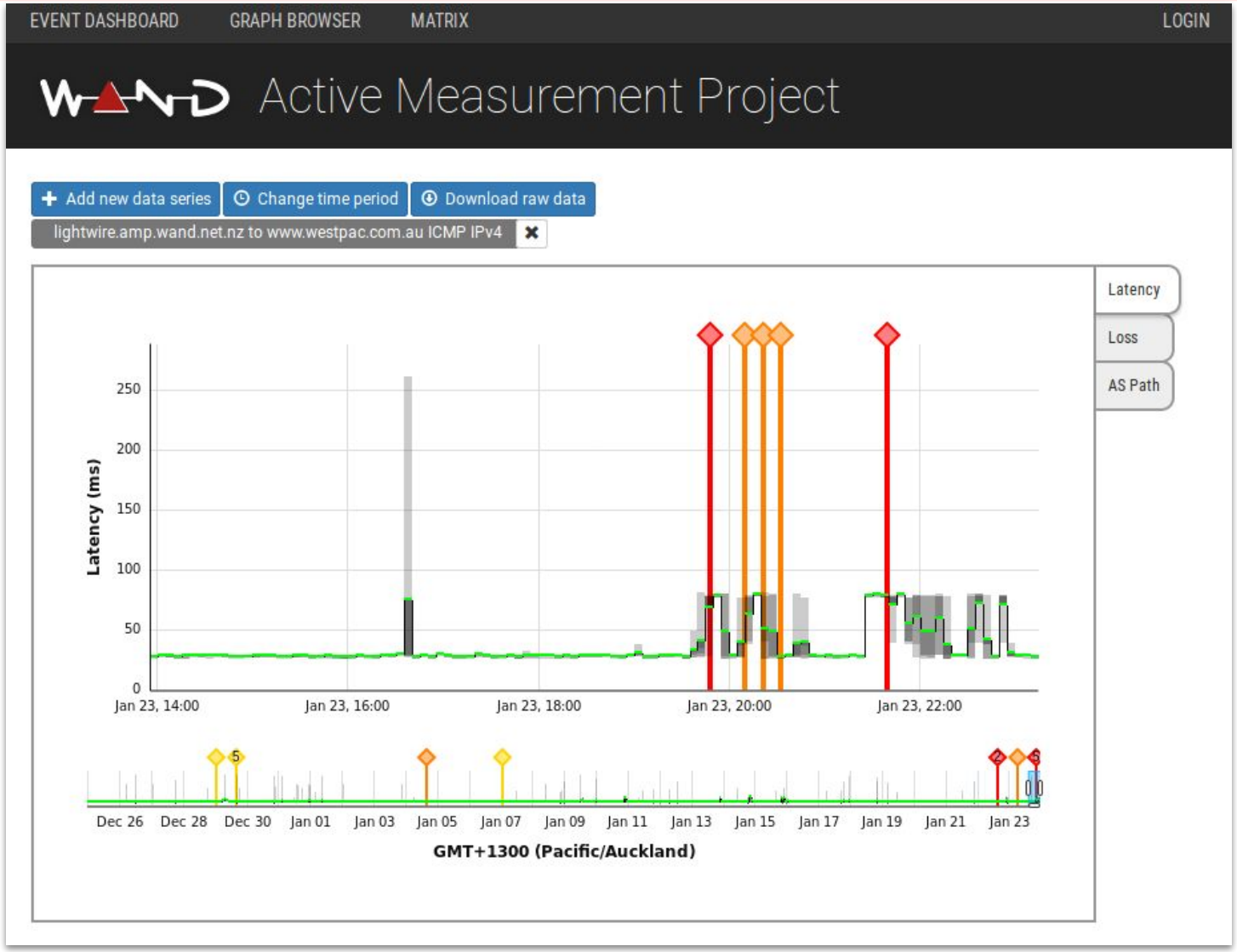
WAND Active Measurement Project

Recent event groups

Change Event Filters

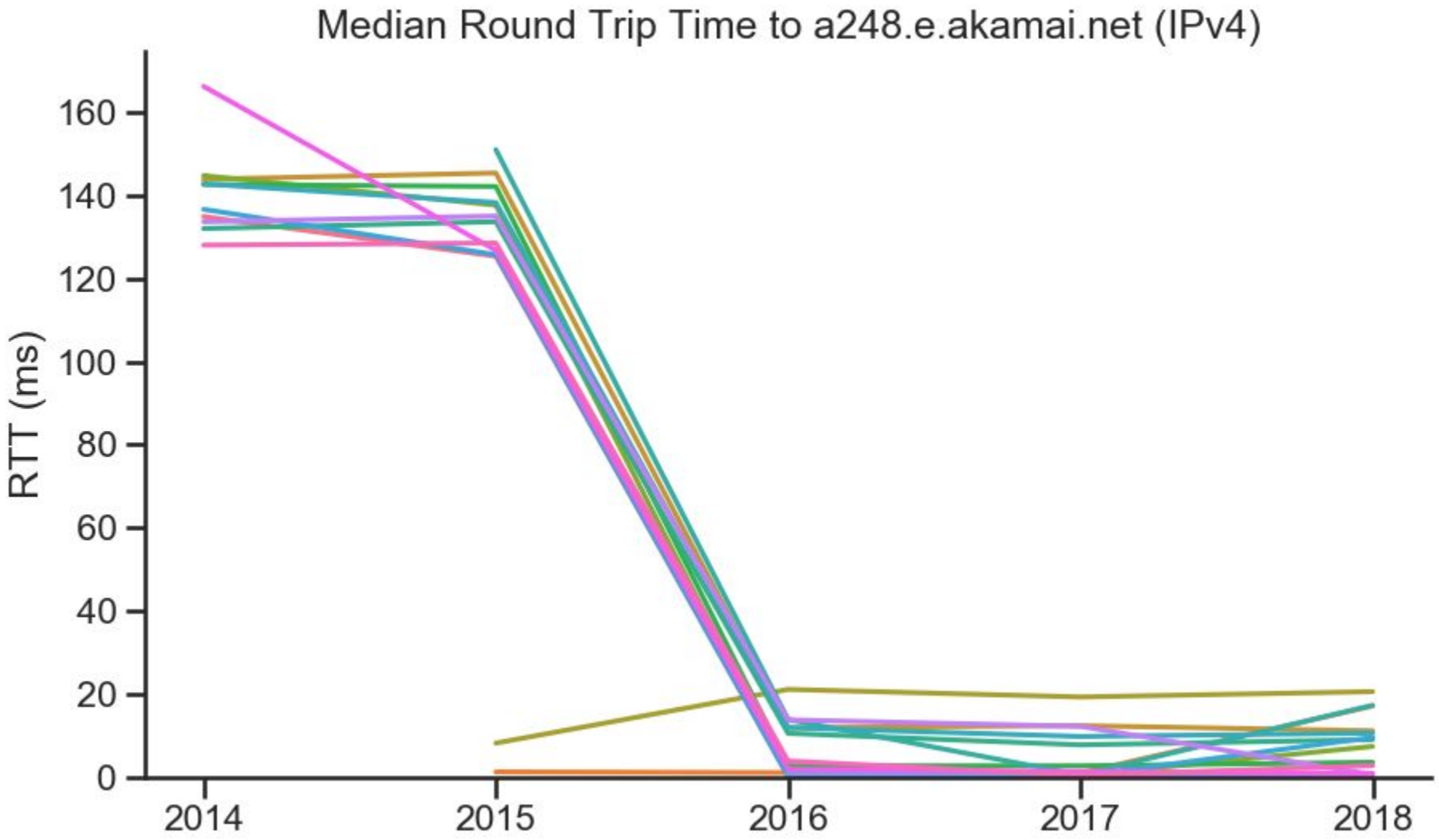
21:33:00 Today	www.kiwibank.co.nz	2 1
21:33:00 Today	www.westpac.com.au	2 1
21:37:00 ICMP latency from lightwire.amp.wand.net.nz to www.westpac.com.au (ipv4) Series has become more variable by a factor of 3.7		
21:33:14 TCP latency from ns3b.amp.wand.net.nz to www.westpac.com.au:443 (ipv4) Average latency increased from 24.51 to 35.77 ms		
21:33:00 ICMP latency from ns3b.amp.wand.net.nz to www.westpac.com.au (ipv4) Mode latency has increased from 23 to 36 ms		
21:32:14 Today	Vibe Communications LTD ASN for anycast peering of	1 3
21:32:14 Today	Vocus Connect International Backbone ICONZ Ltd	1 3
21:32:14 Today	www.chorus.co.nz	4 1
21:32:14 Today	www.imdb.com	3 1

AMP Web Interface



Some Graphs

The Internet Is Getting Closer



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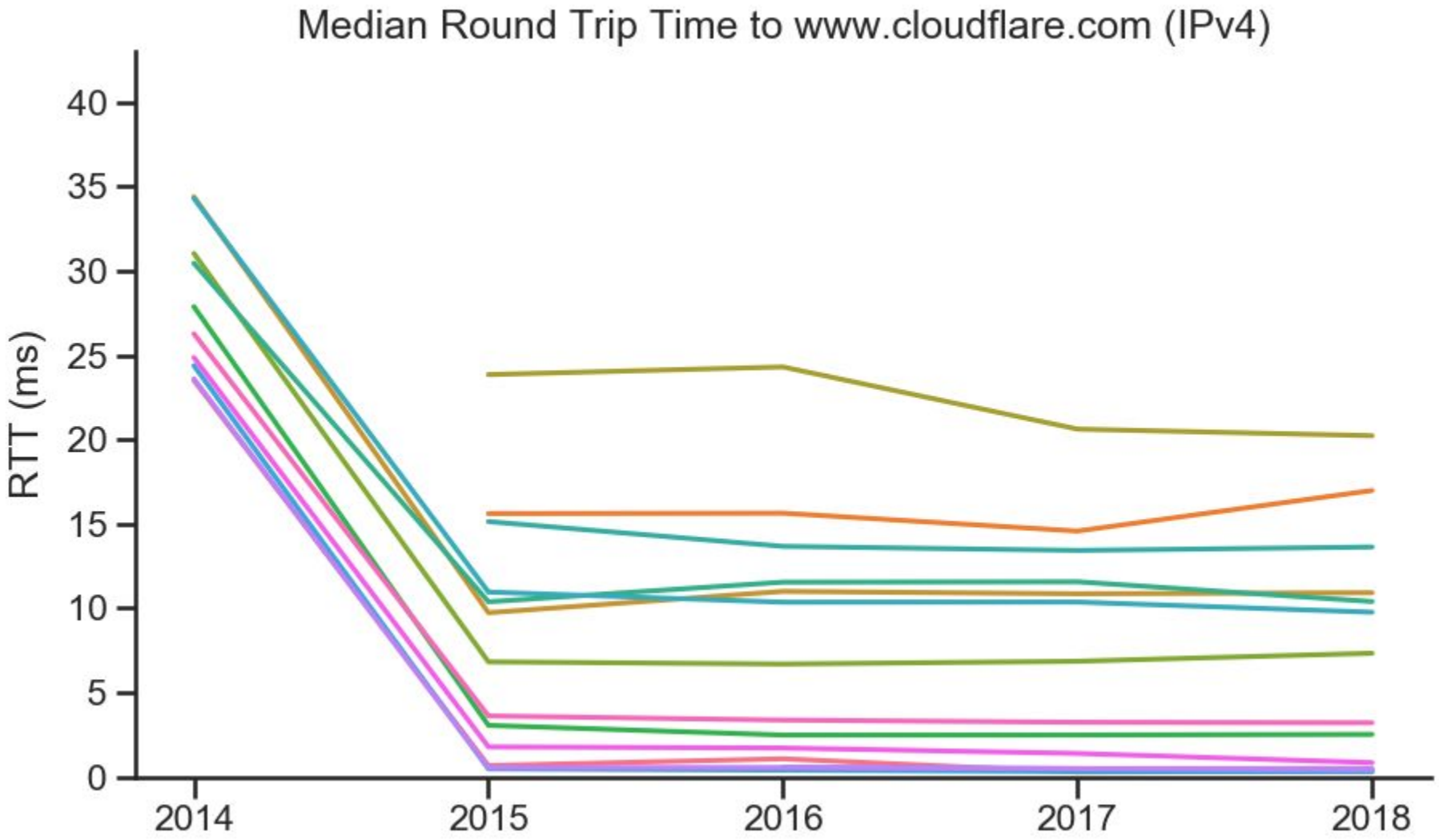

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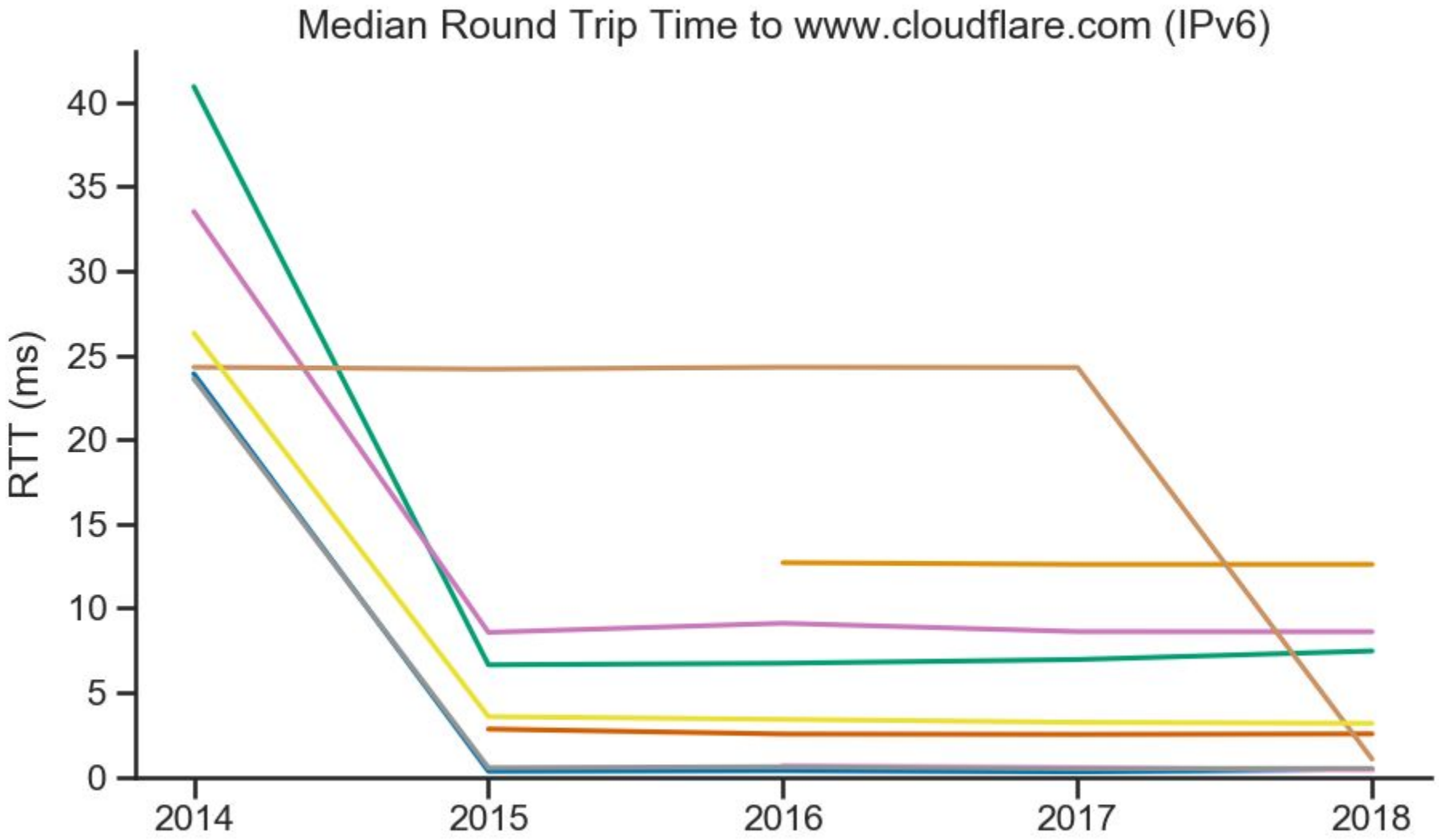
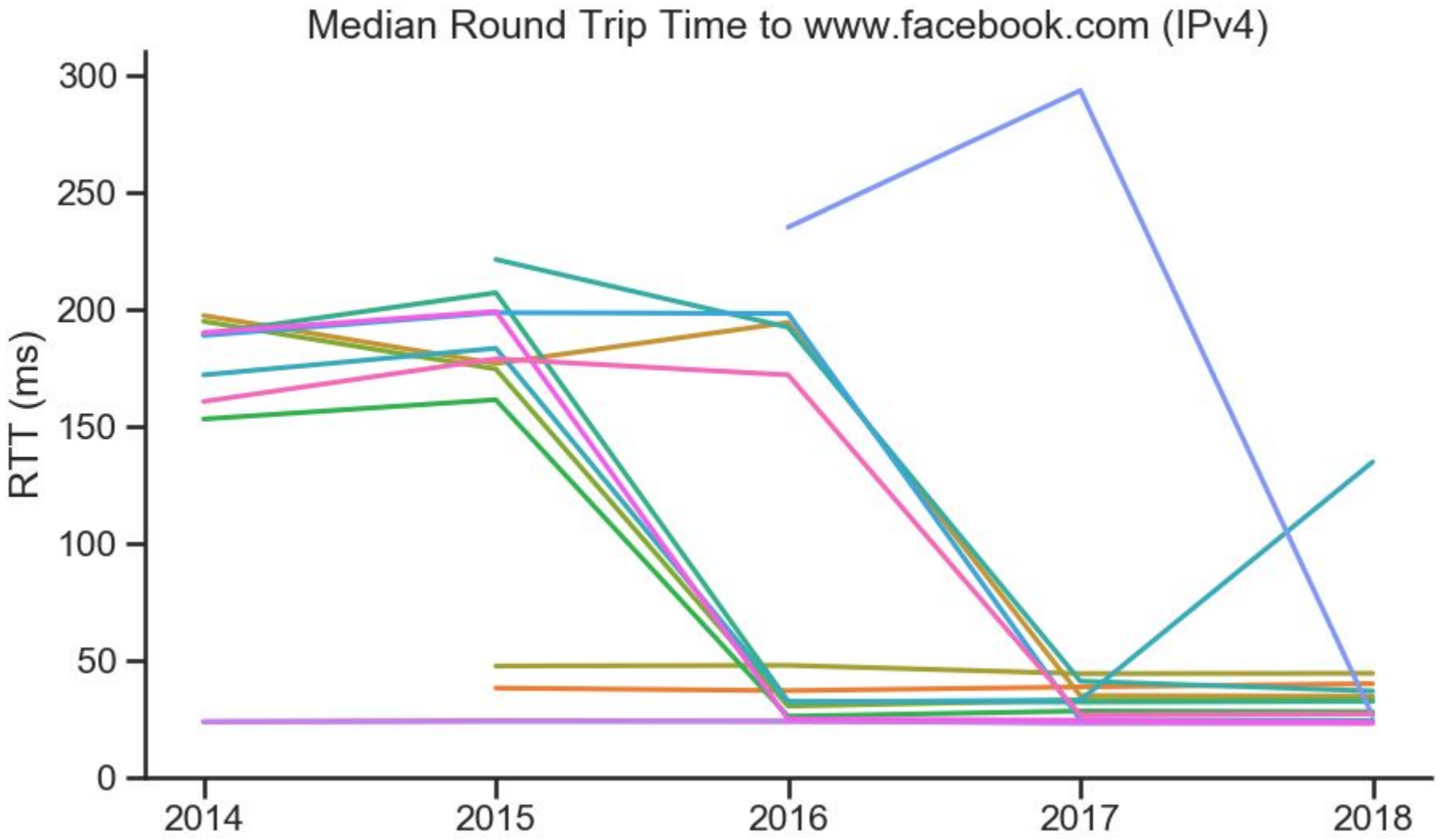


At Least, It's Now In Australia



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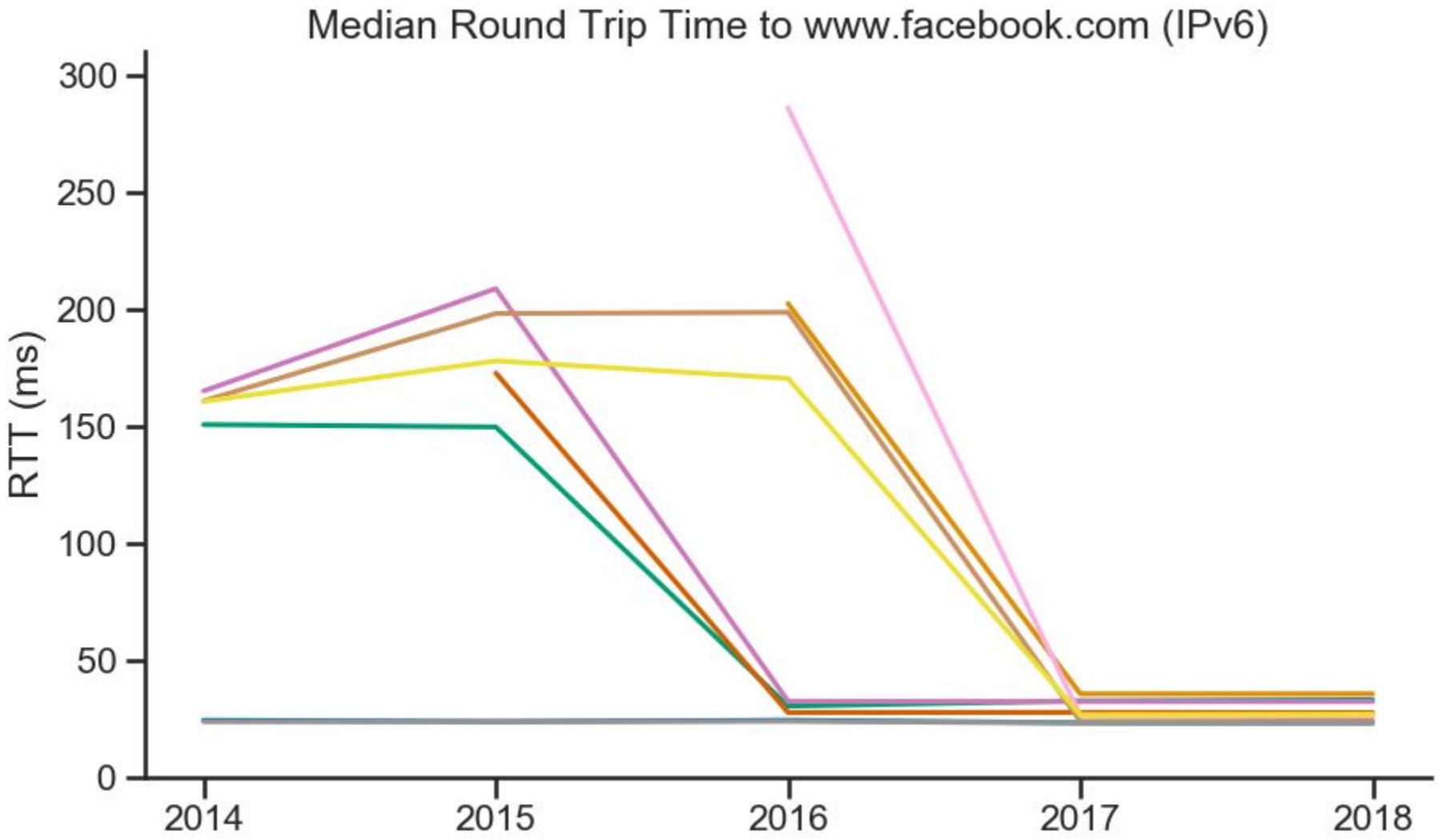
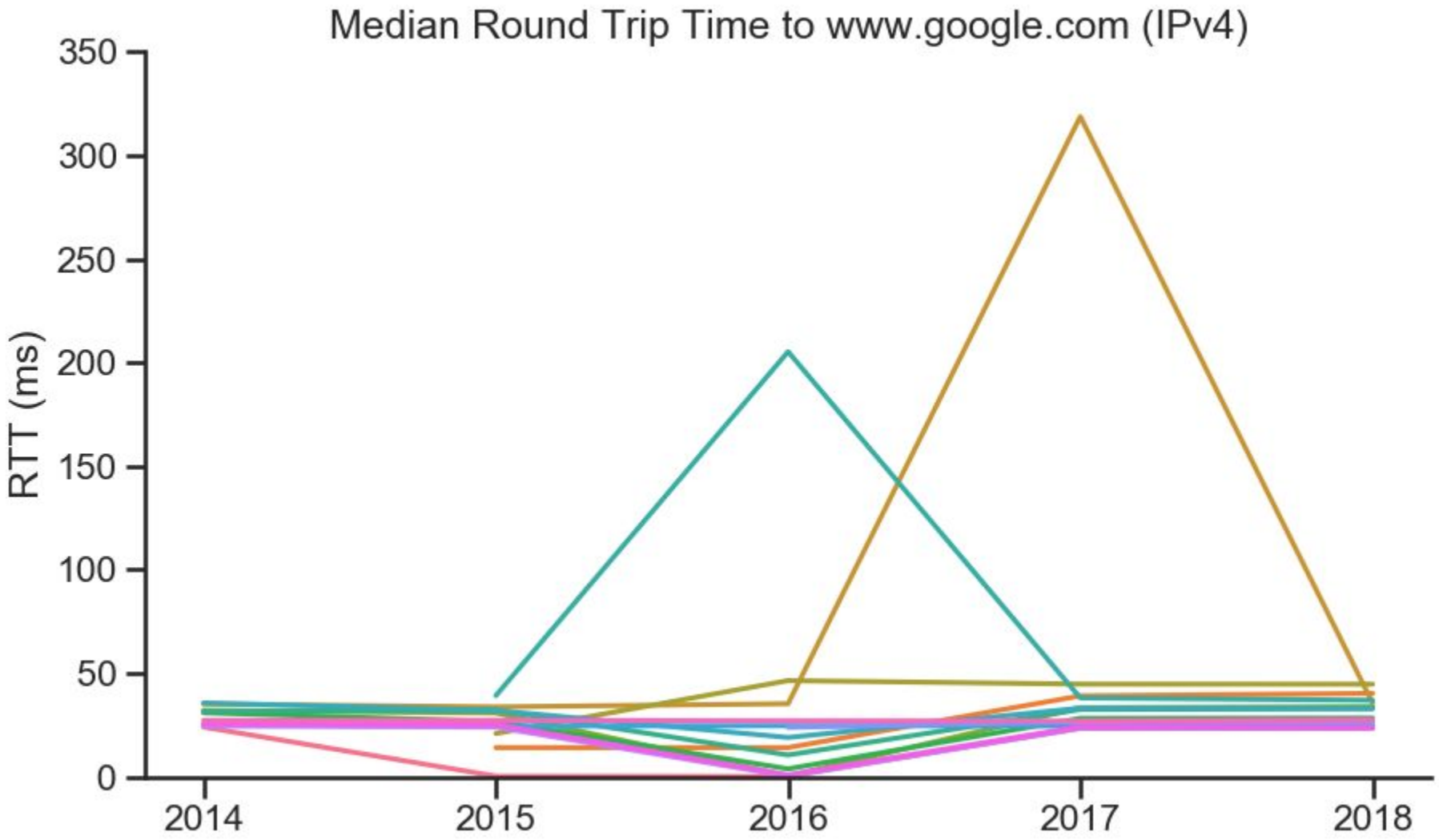


At Least, It's Now In Australia (Usually)



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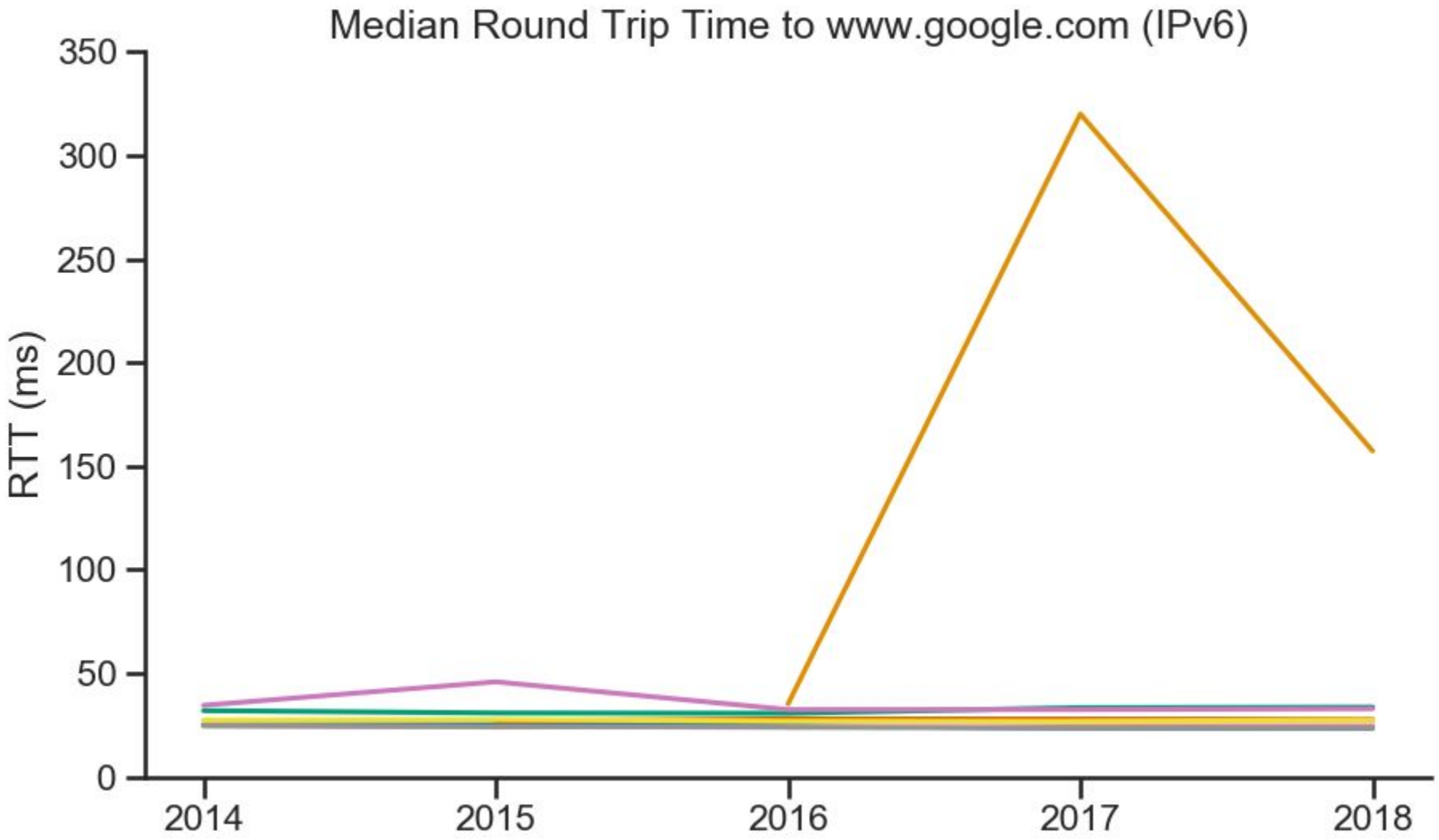
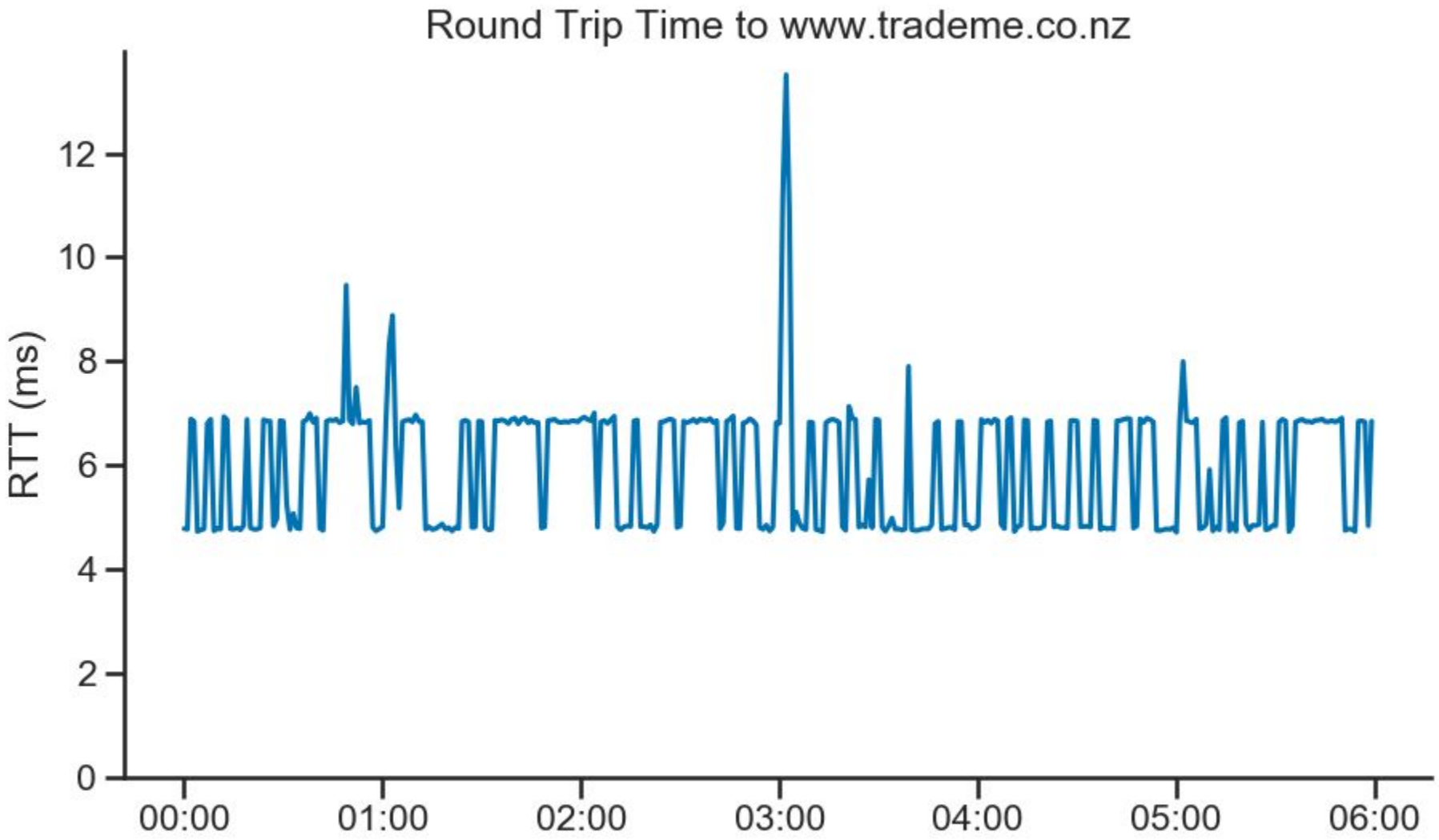
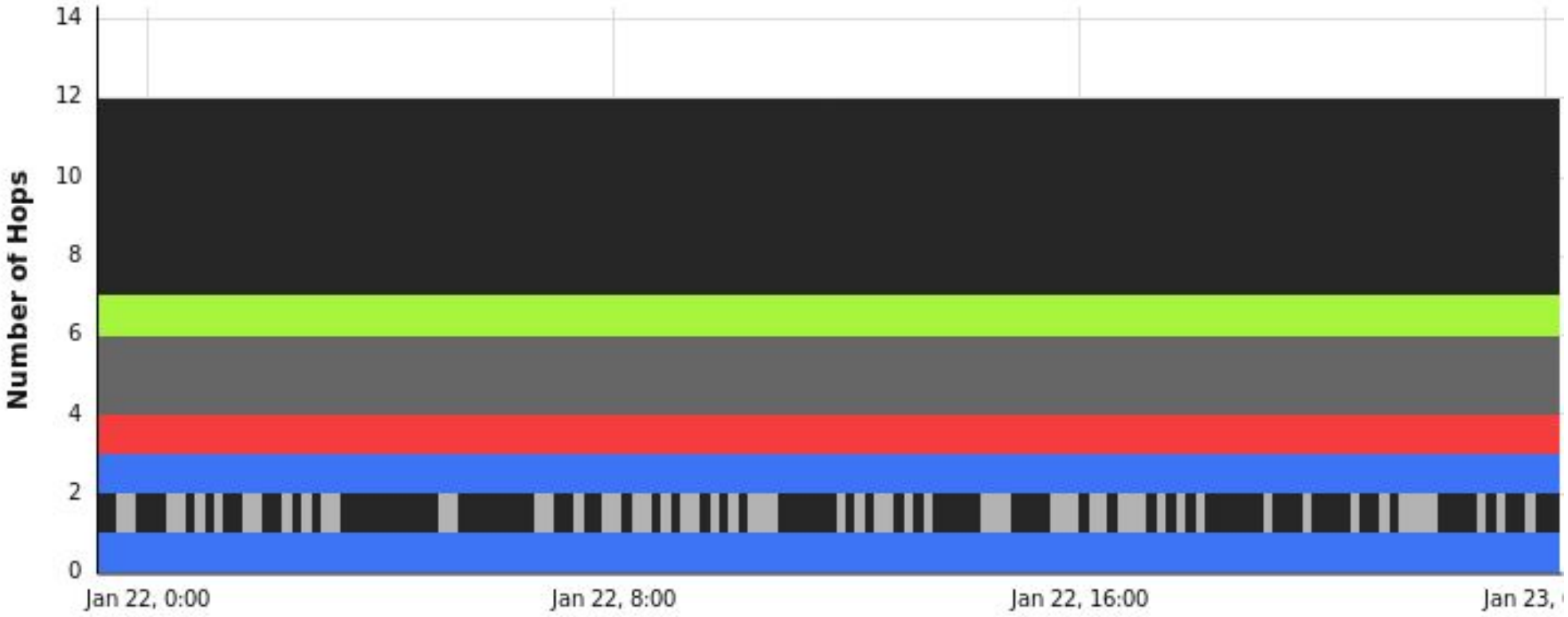


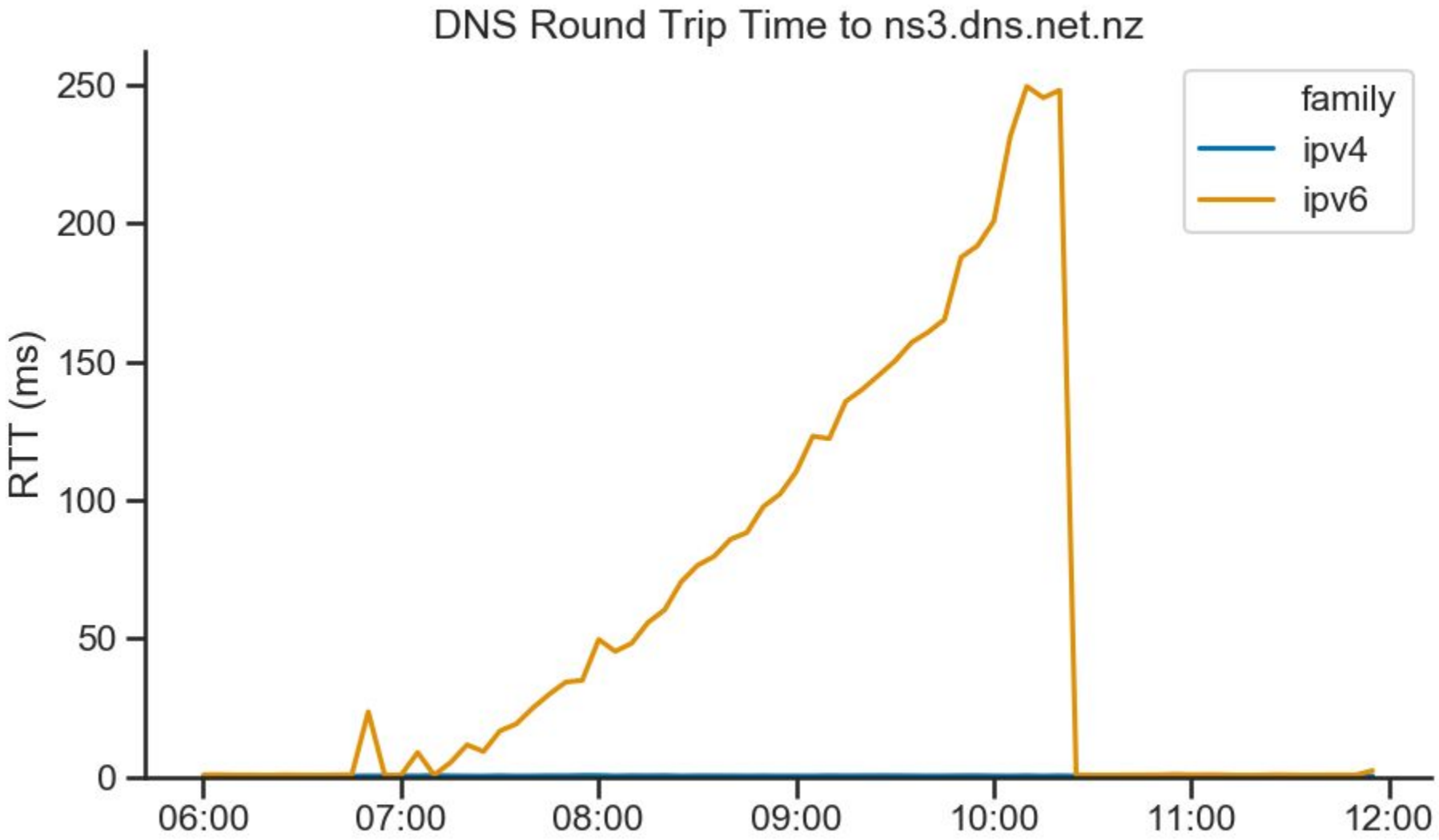
Sometimes Things Move A Bit



Sometimes Things Move A Bit (Or Not)



Odd DNS Response Latency



What Next?

- If you are interested, please get in touch
  - Hosting a probe
  - Running your own AMP network
- I can provide some support for installation and configuration
- What tests would be useful for you?

Questions?

https://github.com/wanduow/amplet2/

https://amp.wand.net.nz

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Web100 and Web10G

- Projects to instrument TCP and expose internals to users
- Linux kernel patches and userland API
- Slowly being added to the mainline kernel

TCP_INFO

- TCP configuration
- Latency estimates
  - Minimum
  - Smoothed value
  - Variance
- Packet and byte counts
- Time spent not sending
  - Due to an empty send buffer
  - Due to a full receive window

Timestamping

- SO_TIMESTAMP
  - Generate timestamps when packets are received by the kernel
  - Don’t need to wait till they reach userspace
- SO_TIMESTAMPING
  - Generate timestamps when packets are sent, received or both
  - Hardware support